



ZOO 1302

Vertebrate Diversity

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Lesson 4

Subphylum Vertebrata

Learning Outcomes

*At the end of the lesson you would be able to
describe the*

- general characteristics of vertebrates.
- adaptations that have guided early vertebrate evolution.
- earliest vertebrates.

Phylum - Chordata divided into three basic groups:

1. Sub phylum - Urochordata (tunicates)

2. Sub phylum - Cephalochordata (lancelets)

3. Sub phylum – Vertebrata (Craniates)



Protochordates

Subphylum Vertebrata (Craniata)

- Subphylum Vertebrata - **monophyletic group** that shares the basic chordate characteristics with the Urochordates and Cephalochordates.
- Taxon - group of organisms
- Clade - a group of organisms believed to comprise all the evolutionary descendants of a common ancestor
- Monophyletic group - is a taxon which forms a clade that consists of an ancestral species and all its descendants.

- vertebrates - get their name from vertebrae - the series of bones that make up the backbone.
- alternative name of the subphylum, - Craniata
- more accurately describes the group since all have a cranium (bony or cartilaginous braincase)
- whereas, the jawless fishes lack vertebrae.

- There are approximately 52,000 species of vertebrates which include the largest organisms ever to live on the Earth.

- Fishes
- Amphibians
- Reptiles
- Birds
- Mammals

Characteristics of Subphylum Vertebrata

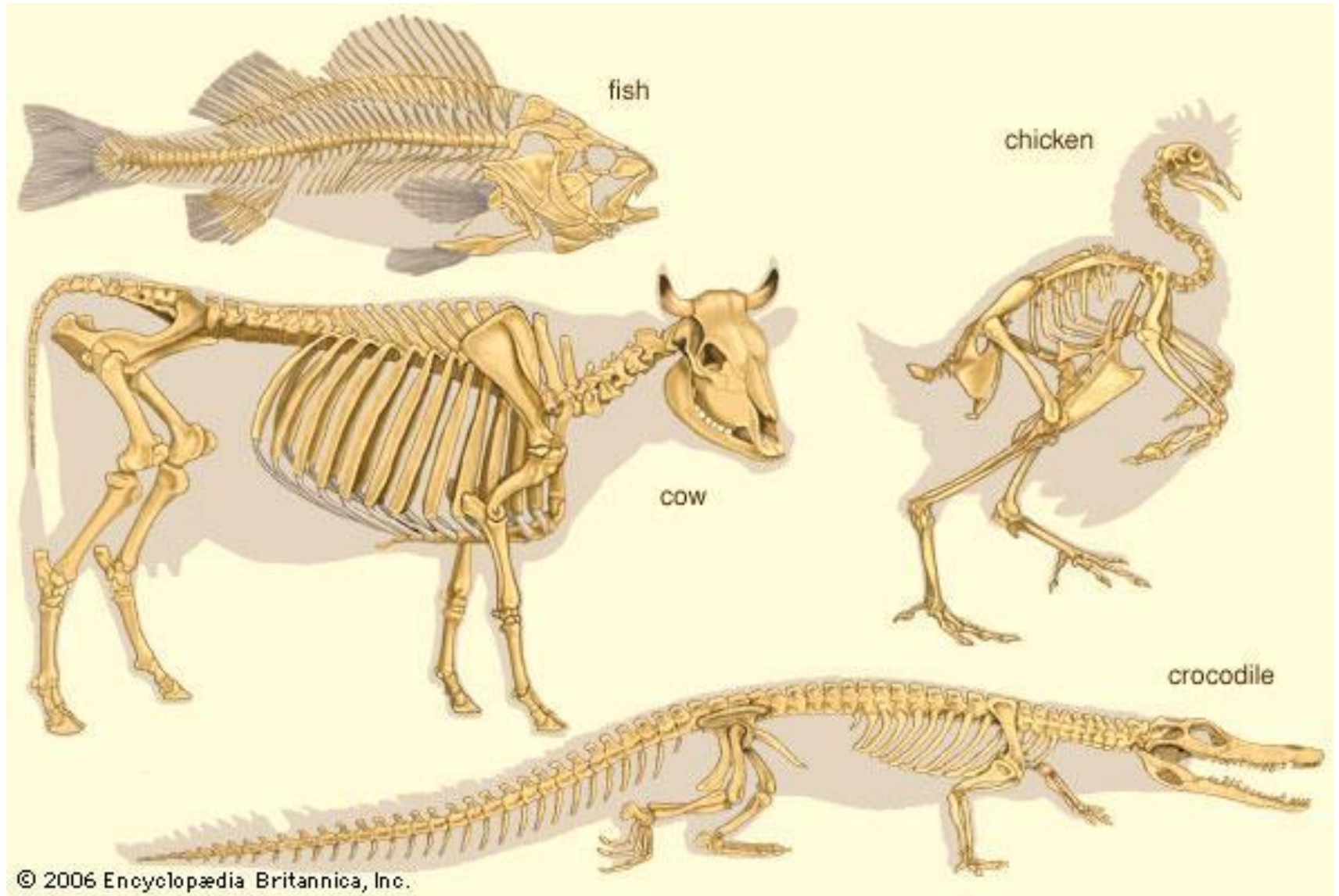
1. All the chief diagnostic features of chordates present at some stage of life cycle

- ✓ Notochord
- ✓ Dorsal tubular nerve cord
- ✓ Pharyngeal pouches
- ✓ Endostyle or thyroid gland
- ✓ Postanal tail

2. Distinctive cartilage or bony endoskeleton consisting

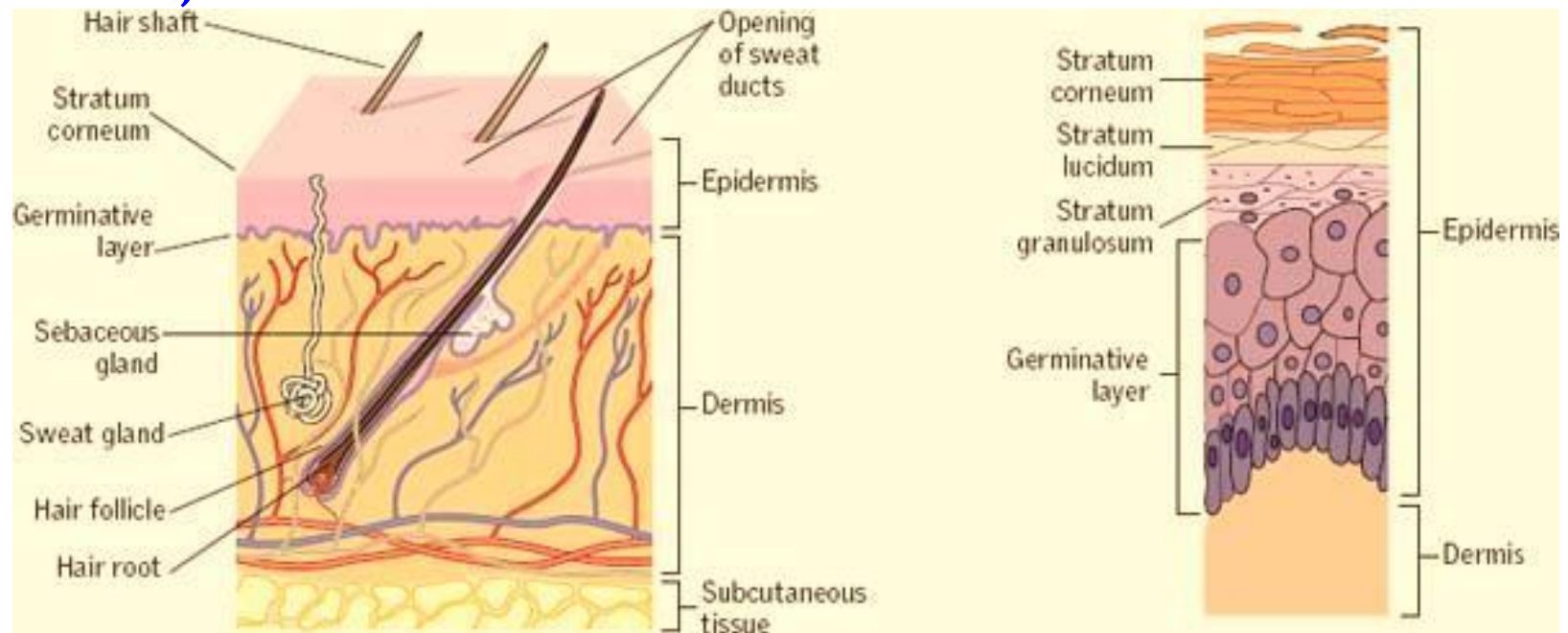
- ✓ **vertebral column** (except hagfishes– lacks vertebrae)
 - to protect the spinal cord
- ✓ **Cranium** – craniata (head skeleton)
 - to protect the brain
- ✓ **Pharyngeal skeleton** – (except eg. Agnathans)
 - the origin of a head opened up a completely new way of feeding for chordates: *active predation*
- ✓ **Two pairs of jointed appendages** – derived from somatic mesoderm
- ✓ Important for muscle attachment

Derived
from
neural
crest
cells



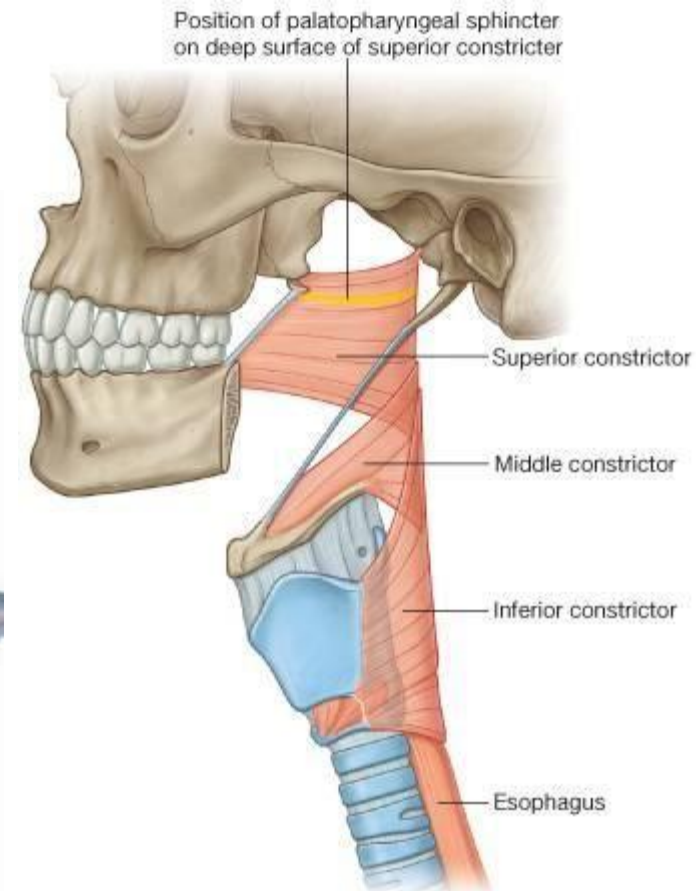
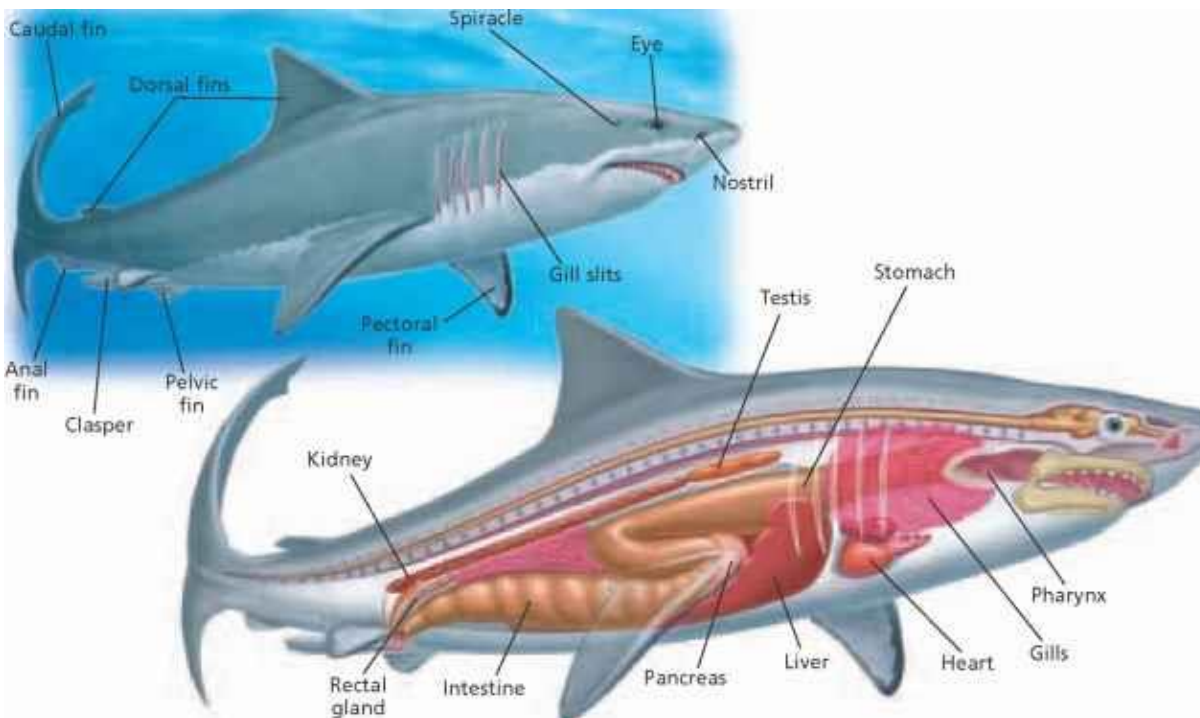
3 . Integument

- two divisions
- an outer epidermis of stratified epithelium from ectoderm
- an inner dermis of connective tissue from mesoderm
- Many modifications of skin eg. glands, scales feathers, claws, horns and hair



4. Muscular pharynx

- In fishes – pharyngeal pouches open to the outside as slits and bear gills
- In tadpole – pharyngeal pouches are source of several glands



5. Myomeres

- Complex, W- shaped muscle segments present
- Provide movement

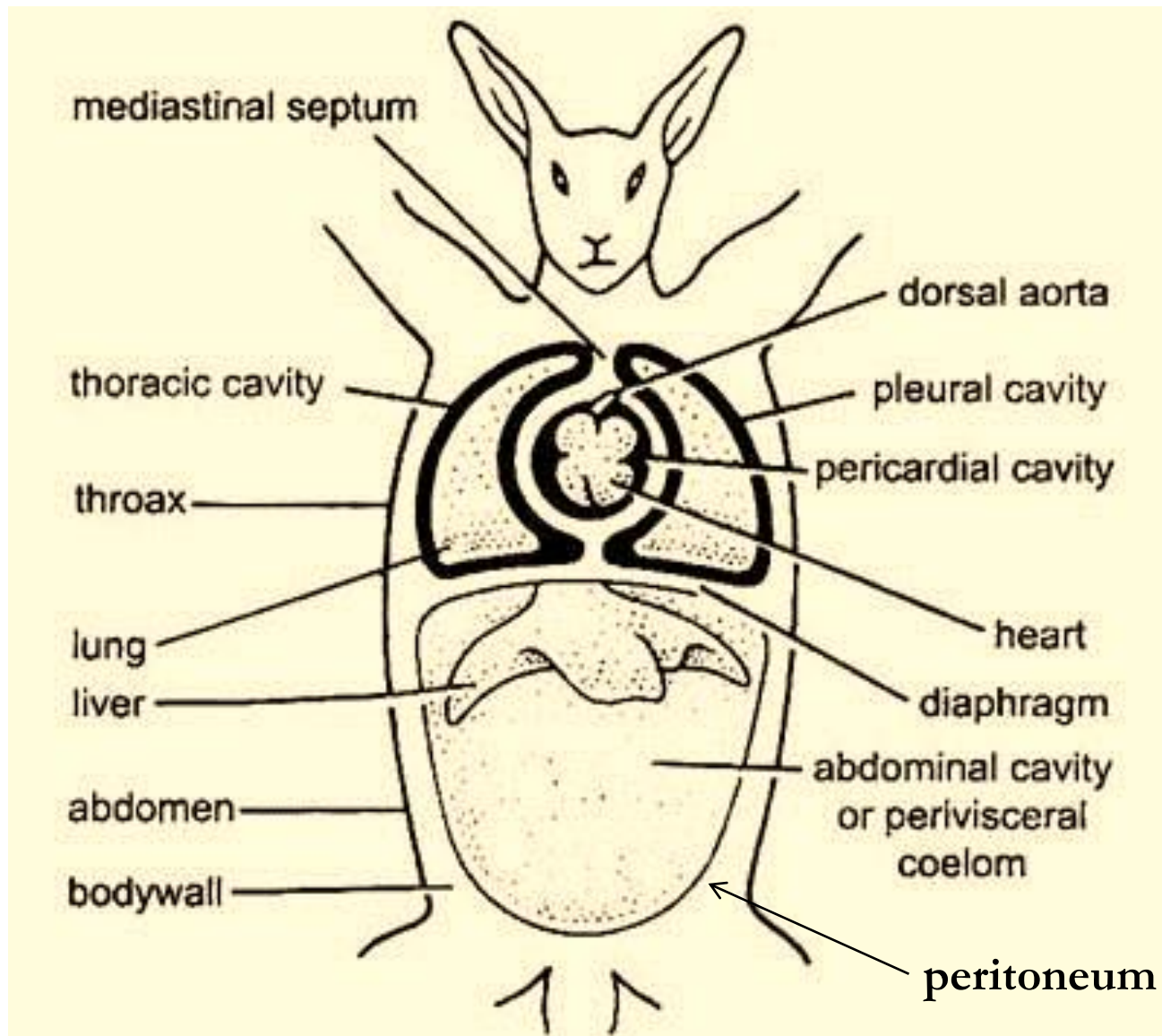


6. Complete muscularized digestive tract

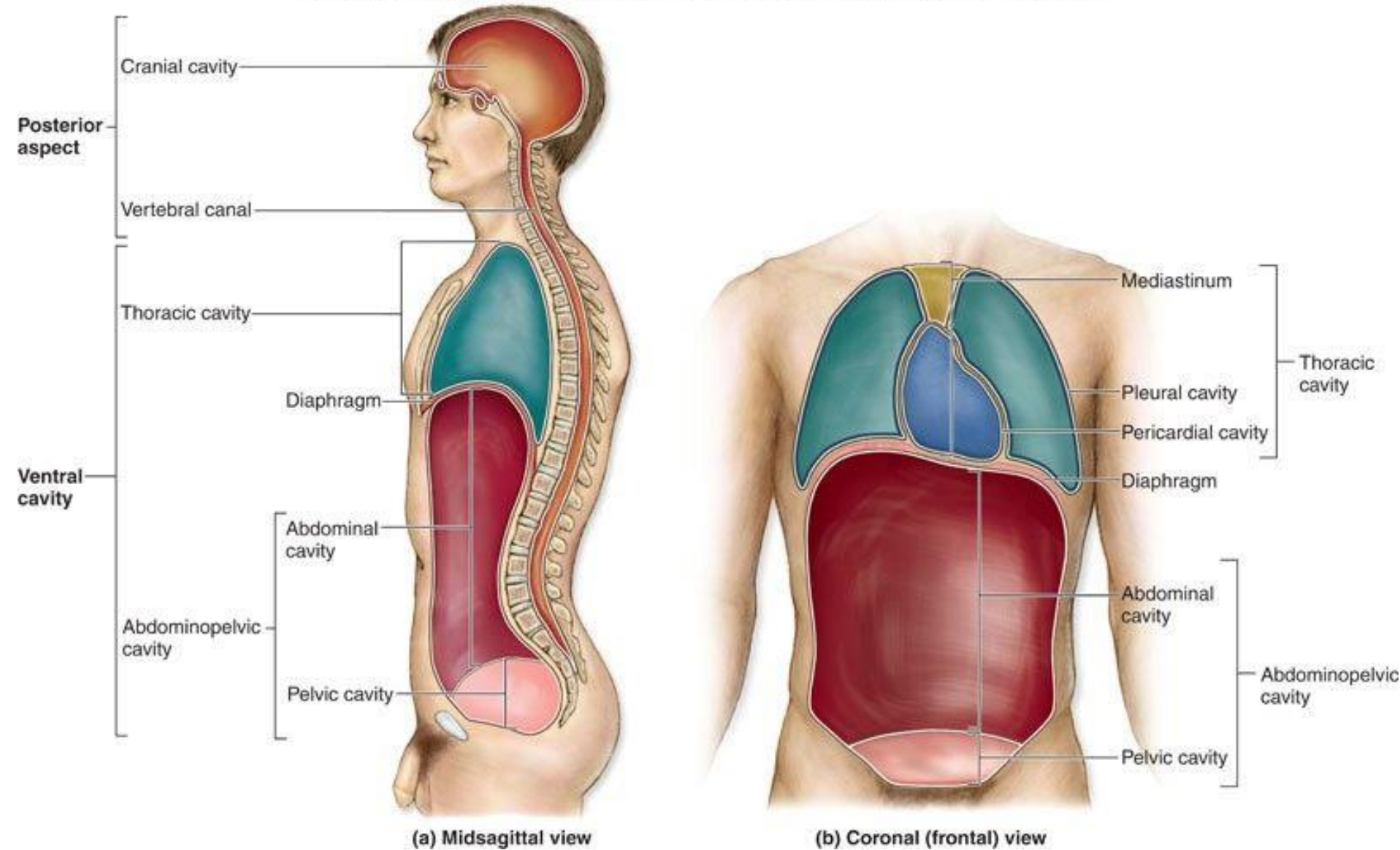
- ventral to the spinal column
- provided with large digestive glands, with distinct liver and pancreas

7. Well-developed coelom

- divided into
 - ✓ pericardial cavity (coelomic cavity around heart)
 - ✓ pleuroperitoneal cavity (coelomic cavity around all visceral organs except heart)
- mammals with a thoracic cavity



Rabbit. Division of coelom or body cavities.



8. Circulatory system

- ventral heart of two to four chambers
- closed blood vessel system of arteries , veins & capillaries
- blood contains erythrocytes with hemoglobin and white corpuscles
- **aquatic vertebrates** – paired aortic arches connecting ventral and dorsal aorta & giving branches to gills
- **Terrestrial vertebrates** – aortic arches modified into pulmonary and systemic circuits

Evolution of circulatory system

Not everyone has a 4-chambered heart



fish

2 chamber



amphibian

3 chamber



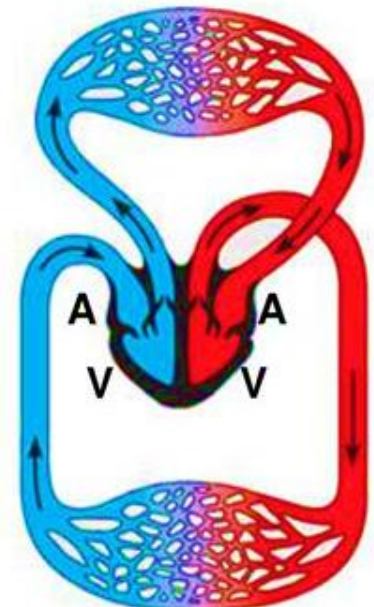
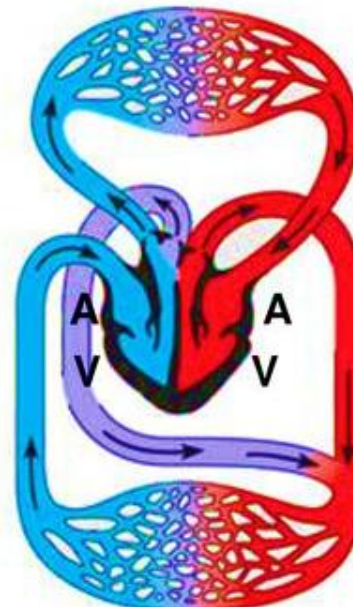
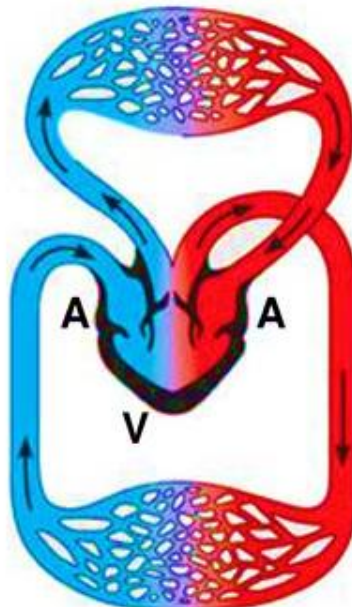
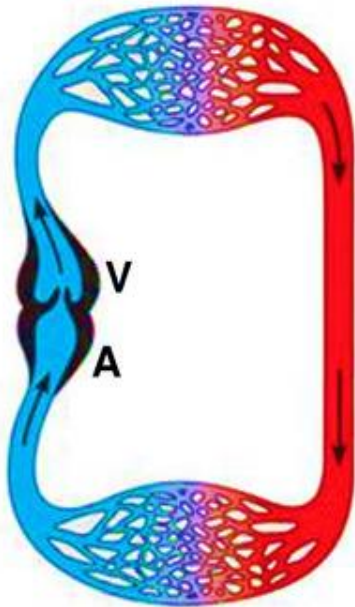
reptiles

3 chamber



birds & mammals

4 chamber



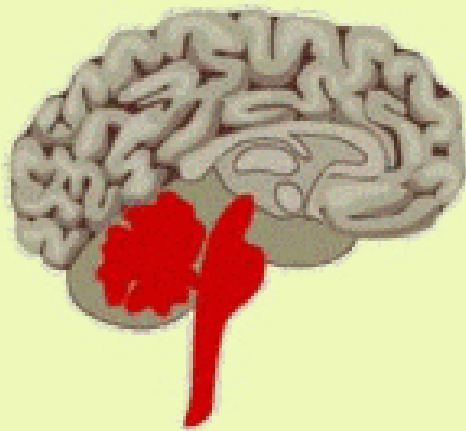
9. Paired special sense organs

- derived from epidermal placodes
 - neurogenic placode - an area of thickening of the epithelium in the embryonic head *ectoderm layer* that gives rise to *neurons* and other structures of the *sensory nervous system*.

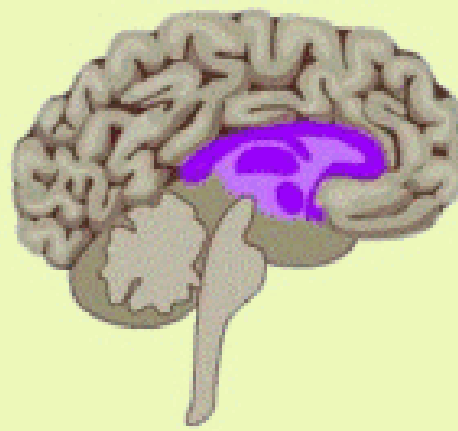
10. Highly differentiated brain (Tripartite brain/Truine brain)

- 10 or 12 pairs of cranial nerves usually with both motor and sensory functions
- A pair of spinal nerves for each primitive myotome
- An automatic nerve system in control of involuntary functions of internal organs

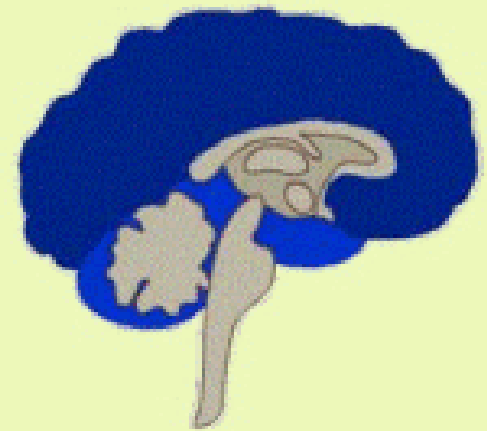
The Three-Parted Brain



Lizard Brain
(Brain stem and cerebellum)
Autopilot
Fight & Flight



Mammal Brain
(Limbic System)
Emotions
Memories
Habits
Attachments



Human Brain
(Neo-Cortex)
Language, abstract thought, imagination, consciousness, reasoning, rationalising

(From Paul D. MacLean's model of the "Triune Brain")

11. Excretory system

- paired, glomerular kidneys (mesonephric or metanephric types in adults)
- kidneys provide with ducts to drain waste to the cloaca

12. Endocrine system

- ductless glands scattered throughout the body

13. Separate sexes

- contain gonads with ducts that discharge their products either into cloaca or into special opening near the anus

14. Two pair of appendages

- supported by limb girdles and appendicular skeleton
 - portion of the skeleton of vertebrates consisting of the bones or cartilage that support the appendages

15. Body plan

- consisting typically of head, trunk, and postanal tail.
- neck present in some, especially terrestrial forms.
- two pairs of appendages usually, although entirely absent in some

Adaptations that have guided early Vertebrate Evolution

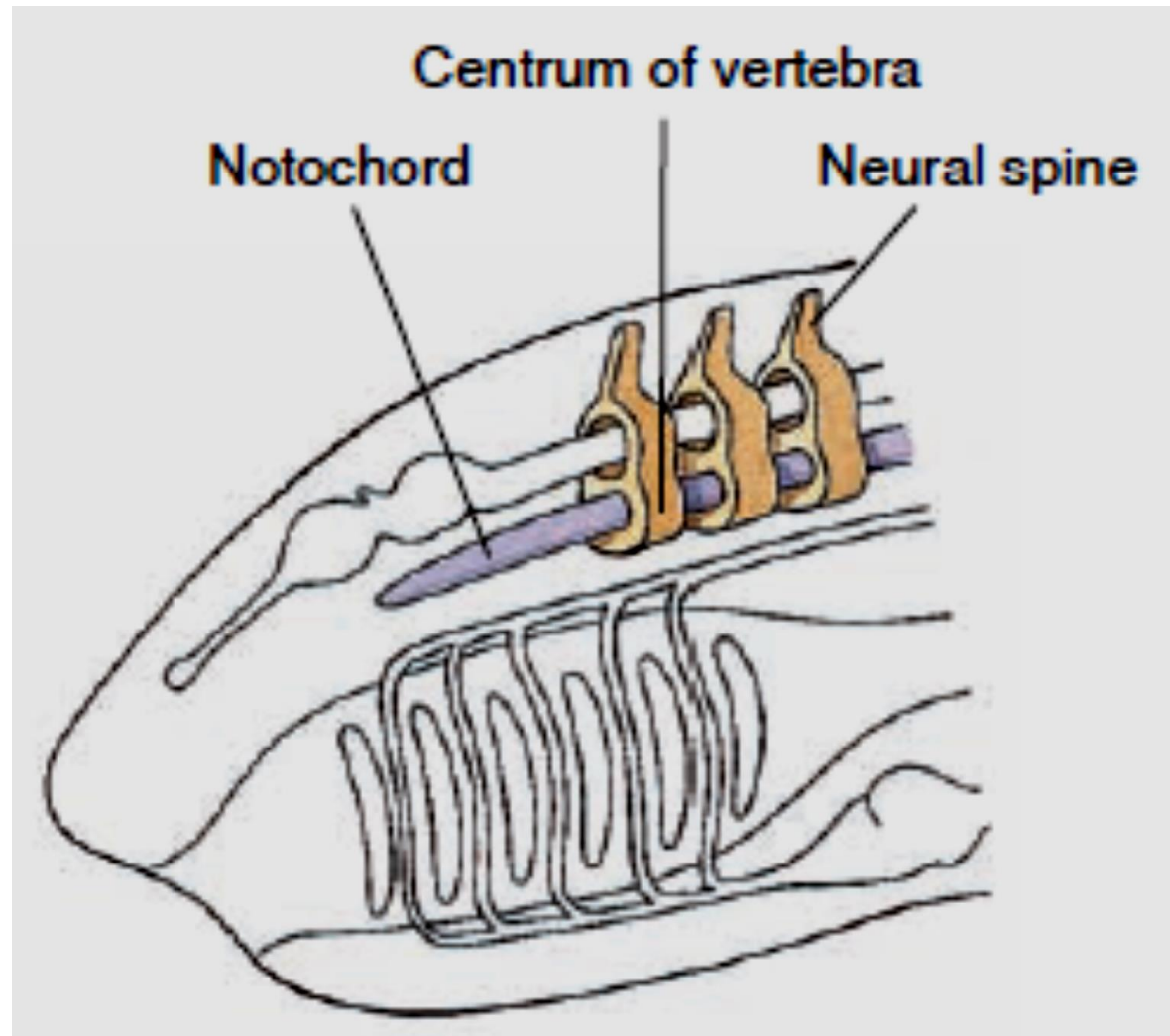
- Earliest vertebrates considerably
 - larger
 - more active than the protochordates.
- The adaptations/modifications
 - a) Musculoskeletal Modifications
 - b) Physiological Upgrade
 - c) Modifications of brain and sensory Systems
 - d) Modifications of neural crest and ectodermal placodes

a) Musculoskeletal Modifications

- Unlimited body size
 - endoskeleton permits almost unlimited body size than the exoskeleton of arthropods.
- Endoskeleton
 - excellent jointed scaffolding for attachment of segmented muscles.
- Segmented body muscles (myomeres)
 - changed from the *V-shaped* muscles of cephalochordates to the *W-shaped* muscles of vertebrates.

- Increased complexity of folding in myomeres
 - provides powerful control over an extended length of the body.
- Presence of fin rays of dermal origin
 - unique to vertebrates aiding in swimming.
- Modifications of skeletal structures and muscles
 - increased speed and mobility
- cartilage, with its fast growth and flexibility
 - construct the first skeletal framework in all vertebrate embryos

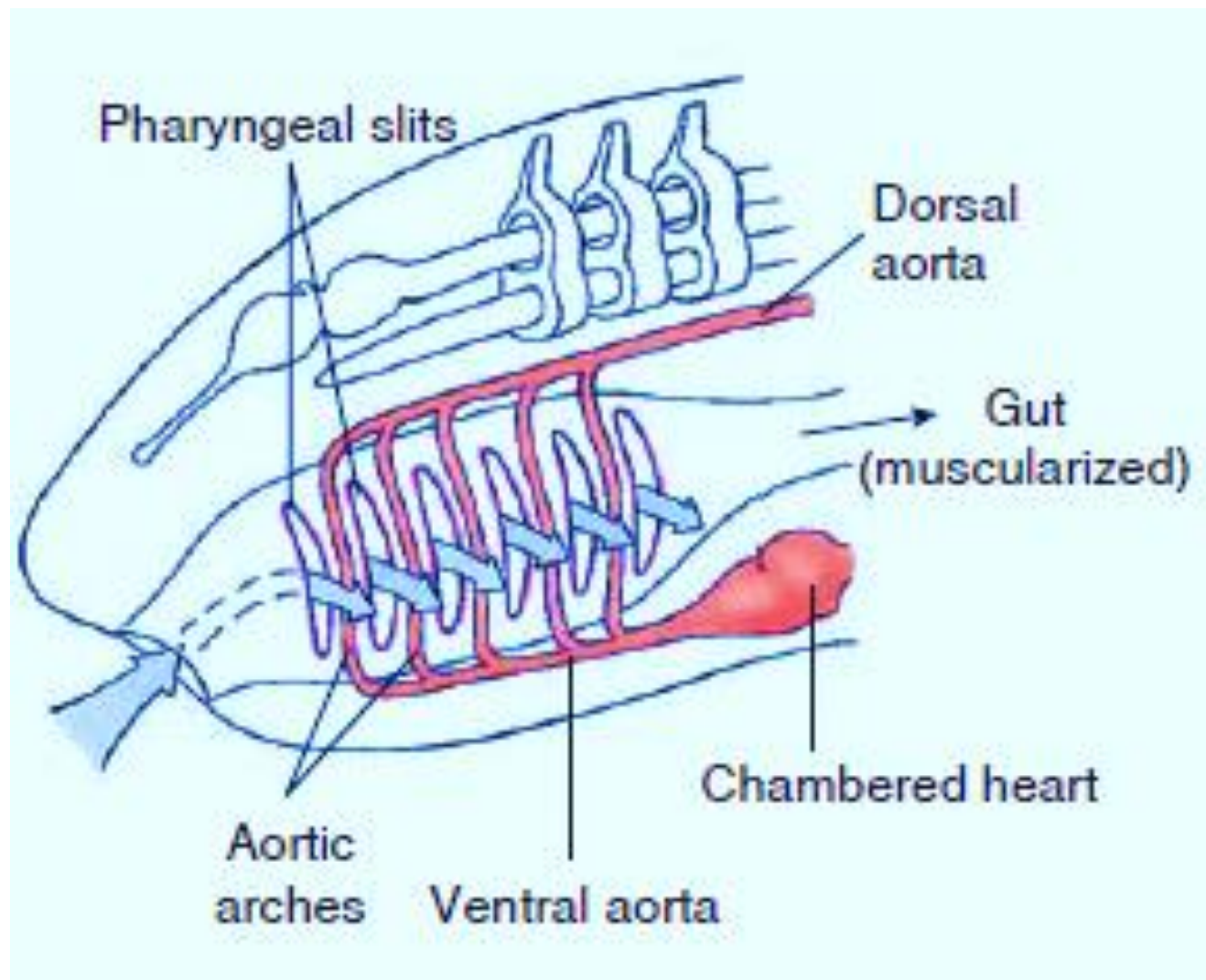
- Scales in fish
 - Some of the most primitive fishes, including ostracoderms and placoderms, were partly covered in a bony, dermal armor.
 - This armor is modified as scales in later fishes.
- Bony encasing of brain
 - develop from cells that originate from the dermis.
- Protection
 - Most vertebrates are further protected with keratinized structures derived from the epidermis, eg. reptilian scales, hair, feathers, claws, and horns.



b) Physiological Upgrade

- High metabolic rate
 - modifications to the digestive, respiratory, circulatory, and excretory systems that meet an increased metabolic demand.
- Perforated pharynx
 - evolved in early chordates as a filter-feeding device, which uses cilia and mucus to move water and to trap small suspended food particles.
 - In vertebrates, the addition of muscles to the pharynx created a powerful pump for moving water.
 - With the origin of highly vascularized gills, the function of the pharynx shifted to *primarily gas exchange*.

- Changes in the gut
 - a shift from movement of food by ciliary action to muscular action
 - addition of *accessory digestive glands*, the liver and pancreas, managed the increased amount of food ingested.
- Ventral three-chambered heart
 - consisting of a sinus venosus, atrium, and ventricle,
 - erythrocytes with hemoglobin enhanced transportation of nutrients, gases, and other substances.



c) Modifications of brain and sensory Systems

■ Brain

- The anterior end of the nerve cord became enlarged as a **tripartite brain** (forebrain, midbrain, and hindbrain)
- protected by a cartilaginous or bony cranium.

■ Sensory Systems

- When vertebrate ancestors shifted from filter-feeding to active predation - new sensory, motor, and integrative controls became essential for *location and capture of larger prey*.

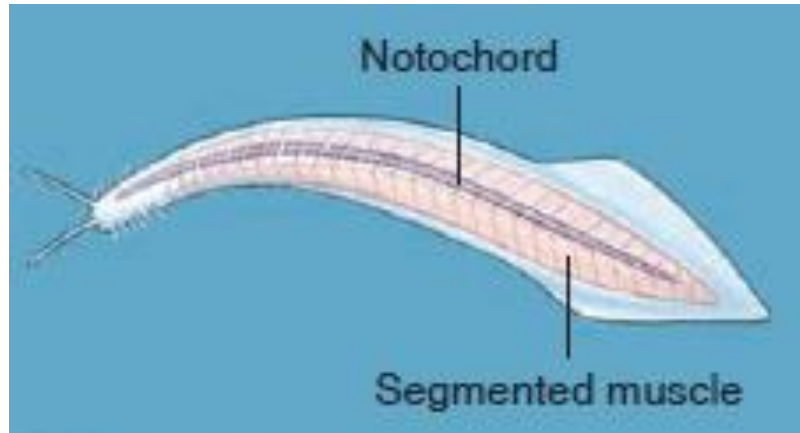
- Sense organs
 - *Paired special sense* organs designed for distance reception evolved.
eg. eyes with lenses and inverted retinas.
 - *Pressure receptors*, such as paired inner ears designed for equilibrium and sound reception
 - *Chemical receptors* including taste
 - Sensitive *olfactory organs*
 - *Lateral-line* receptors for detecting water vibrations
 - *Electroreceptors* for detecting electrical currents that signal prey.

d) Modifications of neural crest and ectodermal placodes

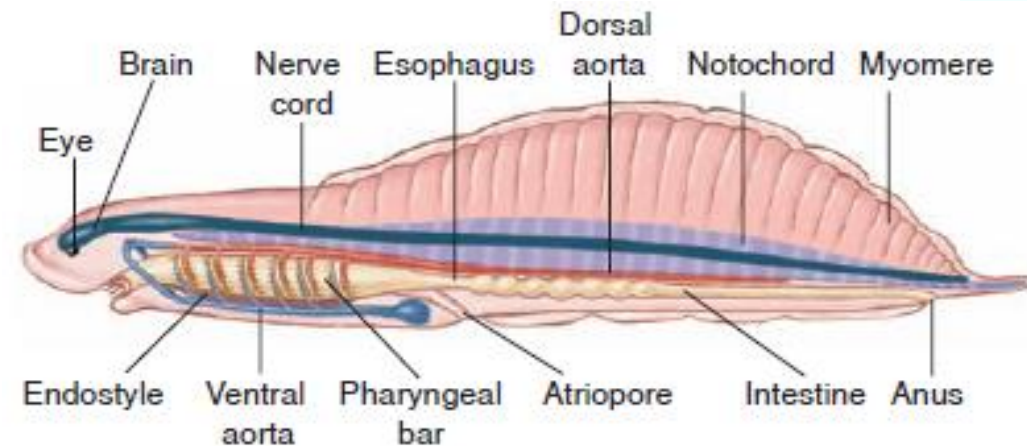
- Neural crest
 - a population of *ectodermal cells* lying along the length of the *embryonic neural tube* contributes to the formation of many different structures.
 - Eg. important structures are cranium, pharyngeal skeleton, tooth dentine, some cranial nerves, ganglia, some endocrine glands, and Schwann cells
 - they may regulate the development of adjacent tissue, eg. tooth enamel and pharyngeal muscles

- Ectodermal Placodes
- Ectodermal placodes (Gr. *placo*, plate) are plate like ectodermal thickenings that appear on either side of the neural tube.
- give rise to the olfactory epithelium, lens of the eye, inner ear epithelium, some ganglia and cranial nerves, lateral-line mechanoreceptors, and electroreceptors.
- placodes also induce the formation of taste buds.
- Thus the vertebrate head with its sensory structures located adjacent to the mouth (later equipped with prey-capturing jaws), stemmed from the creation of new cell types.

Earliest Vertebrates

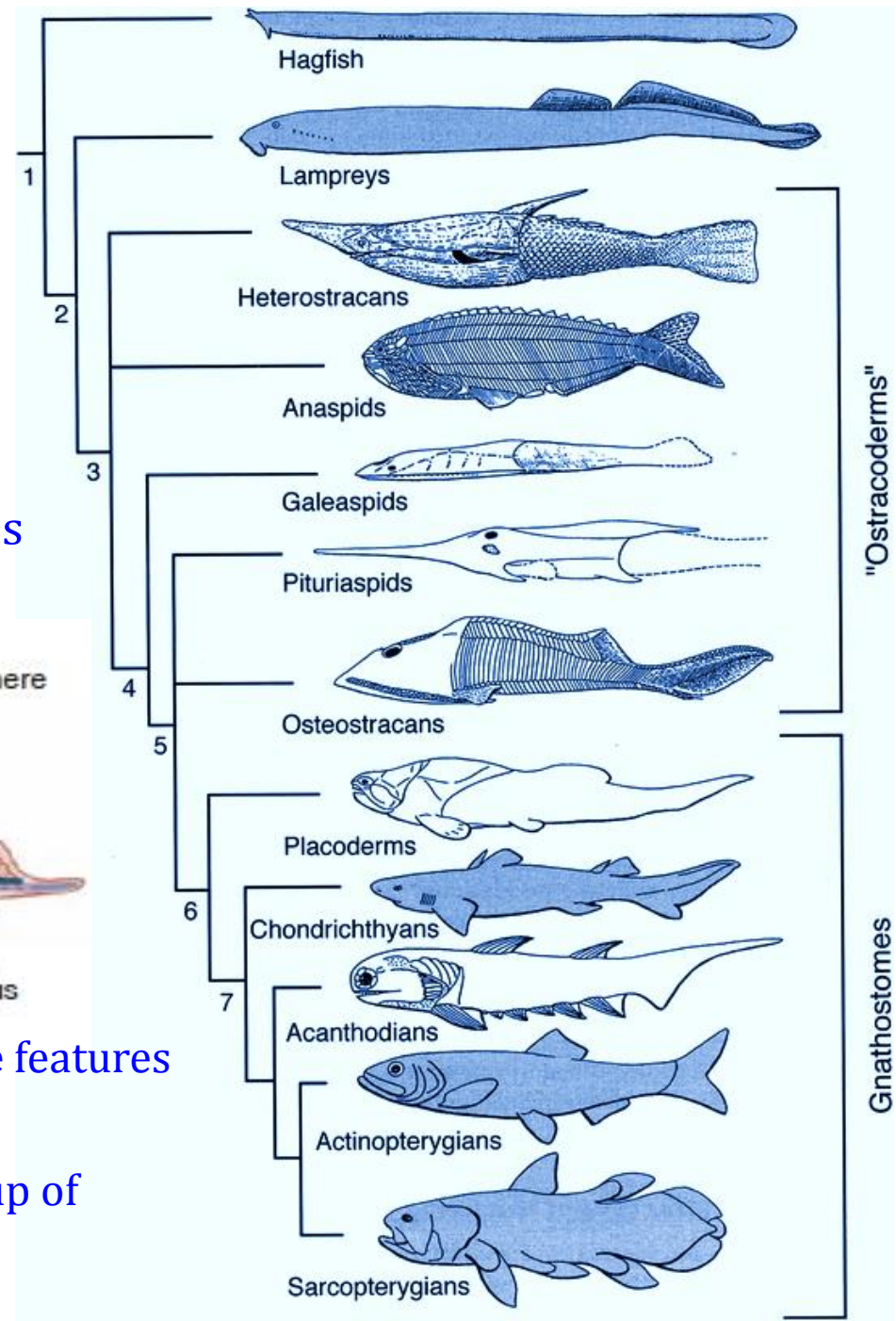


Pikaia, an early chordate from the Burgess Shale of British Columbia, Canada.



Haikouella, a chordate with several vertebrate features from early Cambrian, China.

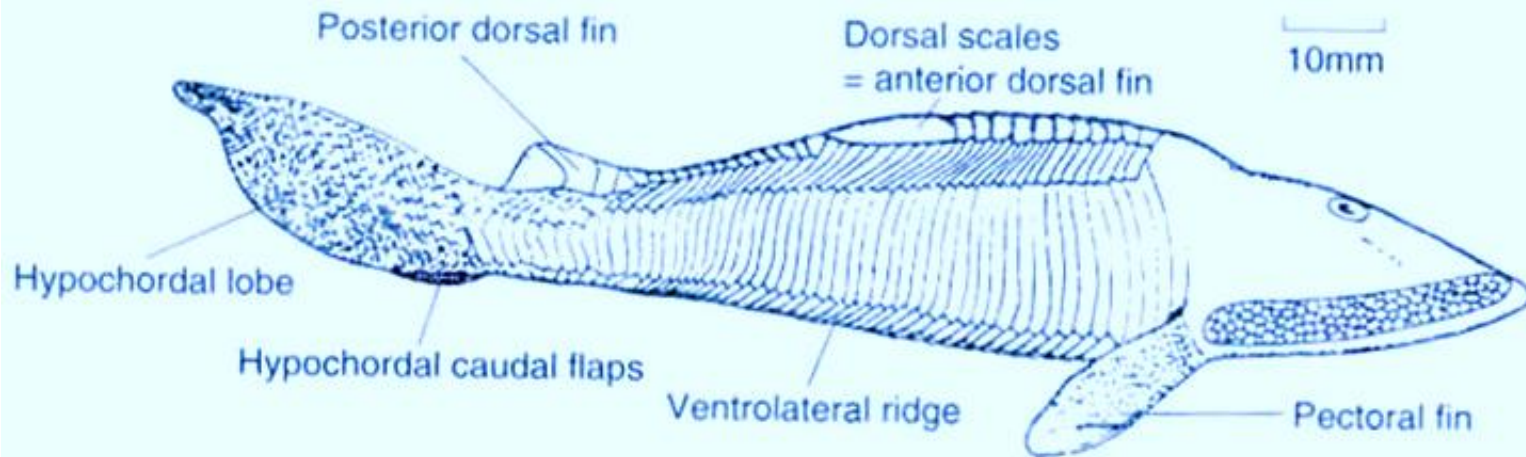
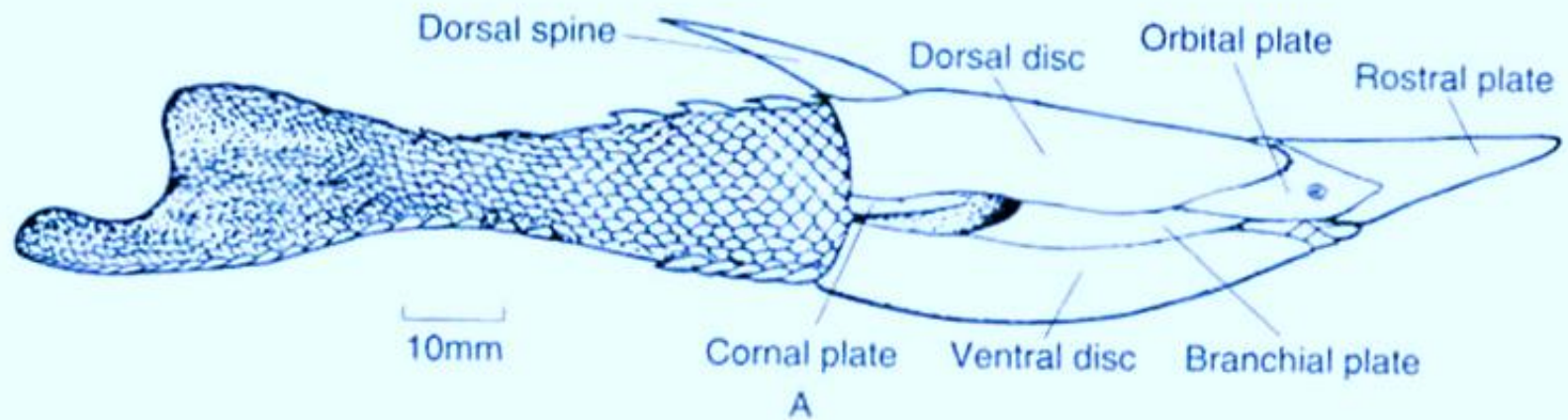
It has been hypothesized to be the sister group of vertebrates (craniates).

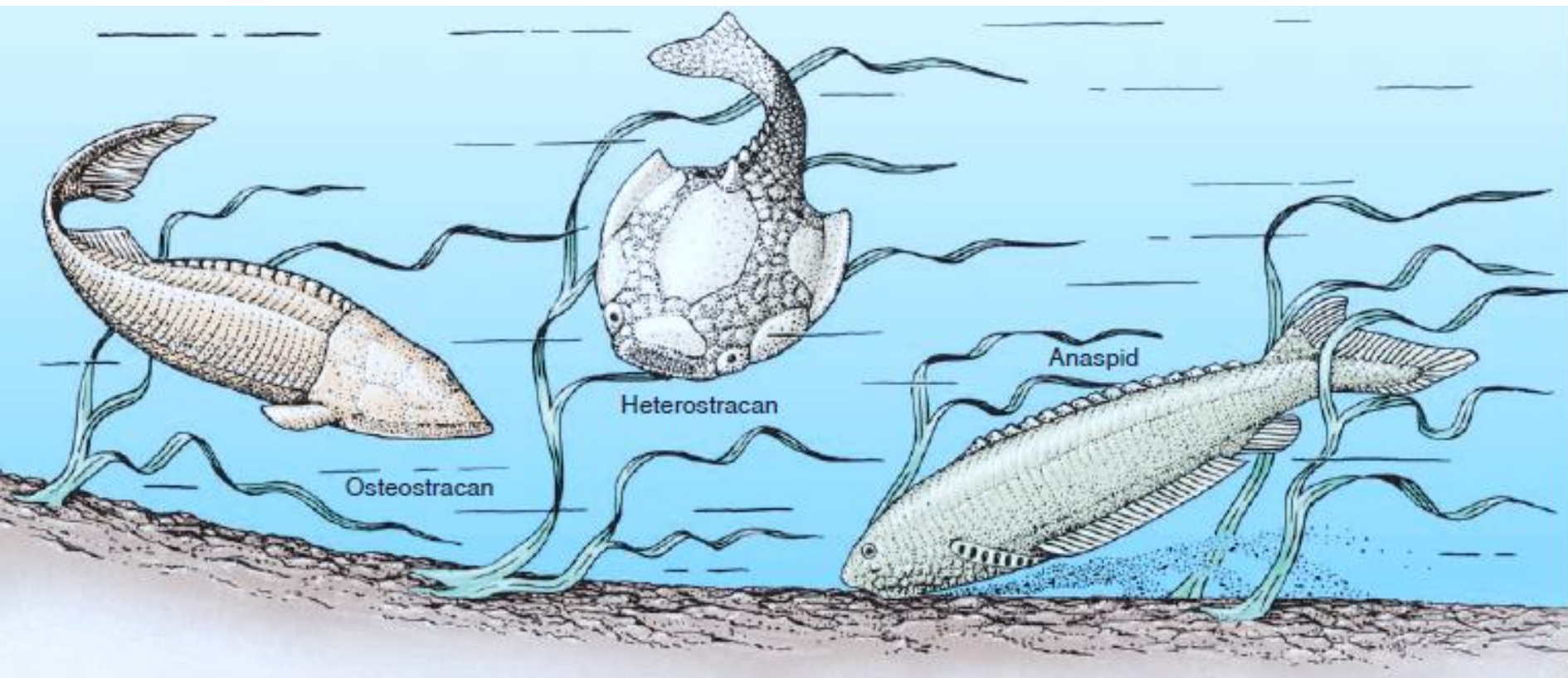


- **Ostracoderms** - earliest known vertebrate fossils, from late Cambrian and Ordovician deposits.
- It was discovered 530-million-year-old fossils in the amazing Chengjiang deposits
 - *Mylokunmingia* - fishlike vertebrates
(Gr. *myllo*, sea fish, Kunming, a city in China)
 - *Haikouichthys*
(Haikou, a city in China, Gr. *ichthy*, fish)
- Hence the origin of vertebrates - early Cambrian.

- These fossils showed many typical vertebrate characters including
 - a heart,
 - paired eyes
 - otic (ear) capsules
 - rudimentary vertebrae

- *earliest ostracoderms were*
 - armored with bone in their dermis
 - lack jaws
 - lacked paired fins - important for stability in later fishes.
 - sucked water into the pharynx by muscular pumping (protochordates - ciliary filter-feeding)
 - active predators fed on soft-bodied animals.





Three ostracoderms, jawless fishes of Silurian and Devonian periods.



The end